

## Section B

- |           |     |                                                                                                                       |          |     |
|-----------|-----|-----------------------------------------------------------------------------------------------------------------------|----------|-----|
| <b>9</b>  | (a) | blocks labelled sensing device / sensor / transducer<br>processor / processing unit / signal conditioning             | B1<br>B1 | [2] |
|           |     |                                                                                                                       |          |     |
| (b)       |     | (i) two LEDs with opposite polarities ( <i>ignore any series resistors</i> )<br>correctly identified as red and green | M1<br>A1 | [2] |
|           |     | (ii) correct polarity for diode to conduct identified<br>hence red LED conducts when input (+)ve or <i>vice versa</i> | M1<br>A0 | [1] |
|           |     |                                                                                                                       |          |     |
| <b>10</b> |     | large / strong (constant) magnetic field                                                                              | B1       |     |